Chapter 7

Diode Applications III: Clippers, Clampers & Multipliers

7.1 Introduction

Beyond basic rectification (AC to DC conversion), diodes are extensively used for **Waveform Shaping**. In agricultural instrumentation, signals from sensors often contain noise or voltage spikes that can damage sensitive microprocessors. We need circuits to modify these signals to fit safe levels or specific shapes.

This lecture covers three essential waveform-shaping circuits:

- **Clippers (Limiters):** Remove or ”clip” a portion of the input signal (e.g., cutting off noise spikes).
- **Clampers (DC Restorers):** Shift the entire signal up or down to a new DC level without changing its shape.
- **Voltage Multipliers:** Generate high DC voltages from low AC inputs (e.g., for electric fence chargers).

Learning Objectives

By the end of this chapter, you should be able to:

- Distinguish between Series and Shunt Clippers.
- Analyze Biased Clipper circuits and draw output waveforms.
- Explain the operation of Positive and Negative Clampers.
- Calculate the peak output voltage of a Voltage Doubler.
- Design a protection circuit for a microcontroller input.

7.2 Clipper Circuits

A clipper circuit uses a diode to ”clip” off a portion of the input signal voltage above or below a certain level. They are also known as **Voltage Limiters**.

There are two main configurations:

1. **Series Clipper:** The diode is in series with the load.
2. **Shunt (Parallel) Clipper:** The diode is in parallel with the load.

7.2.1 Series Clippers

This is essentially a Half-Wave Rectifier.

A. Negative Series Clipper The diode points towards the load.

- **Positive Cycle:** Anode is positive. Diode ON. $V_{out} = V_{in} - 0.7V$.
- **Negative Cycle:** Anode is negative. Diode OFF (Open). $V_{out} = 0$.
- **Result:** The negative half is removed.

7.2.2 Shunt (Parallel) Clippers

Here, the diode is placed in parallel with the load. A series resistor (R) is required to limit current when the diode conducts.

A. Positive Shunt Clipper The diode points **down** (Anode to ground).

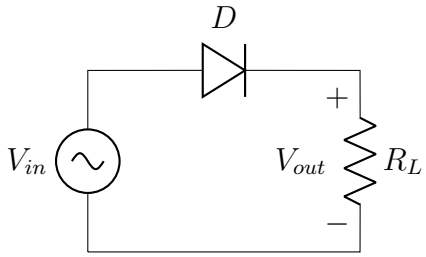


Figure 7.1: Negative Series Clipper Circuit.

- **Positive Cycle:** Anode is positive. Diode turns ON (Short Circuit).

$$V_{out} = V_{diode} \approx 0.7 \text{ V}$$

  The output is clipped at 0.7V.

- **Negative Cycle:** Anode is negative. Diode turns OFF (Open Circuit).

$$V_{out} = \frac{R_L}{R + R_L} V_{in} \approx V_{in} \quad (\text{if } R_L \gg R)$$

  The negative cycle passes through.

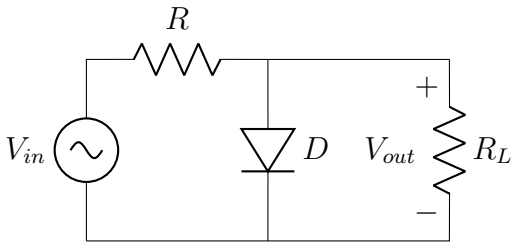


Figure 7.2: Positive Shunt Clipper. Output is clipped at +0.7V.

7.2.3 Biased Clippers

What if we want to limit the voltage to +5V instead of +0.7V? We add a DC voltage source (V_{bias}) in series with the diode.

Operation of Biased Shunt Clipper:

- The diode will only conduct when the input voltage exceeds $(V_{bias} + 0.7V)$.
- **Case 1 ($V_{in} < V_{bias} + 0.7$):** Diode is Reverse Biased (Open). Output follows Input.

- **Case 2 ($V_{in} > V_{bias} + 0.7$):** Diode is Forward Biased (Short). The output is clamped to the battery voltage.

$$V_{out} = V_{bias} + 0.7 \text{ V}$$

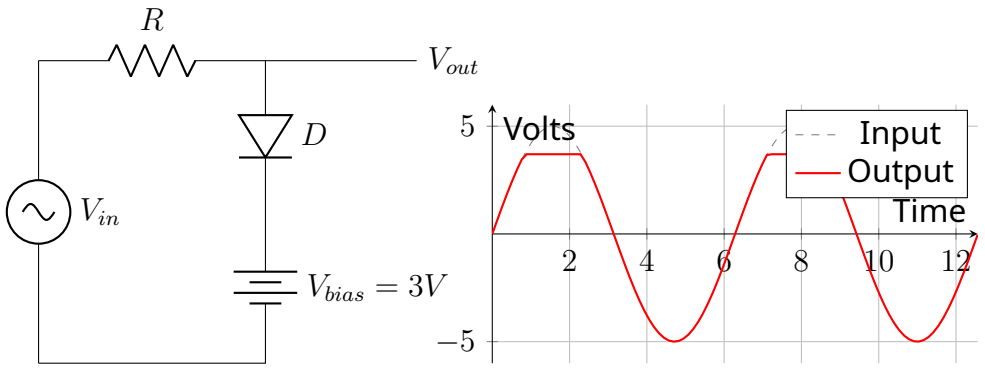


Figure 7.3: Biased Shunt Clipper circuit and waveforms (Clipped at $V_{bias} + 0.7V$).

7.3 Clamper Circuits

A **Clamper** adds a DC level to an AC voltage. Unlike a clipper, it does *not* chop off the signal; it shifts the whole waveform up or down. Ideally, the peak-to-peak voltage remains unchanged.

Essential Components:

1. Diode (Acts as a switch).
2. Capacitor (Stores charge to provide the DC shift).
3. Resistor (Provides a discharge path time constant).

7.3.1 Positive Clamper

This circuit shifts the entire waveform **UPWARDS** so that the negative peak sits at zero (or slightly below).

Analysis:

1. **Negative Half Cycle:** The input goes negative. The diode turns ON (Forward Bias).
   - The capacitor charges rapidly to the peak input voltage V_m.

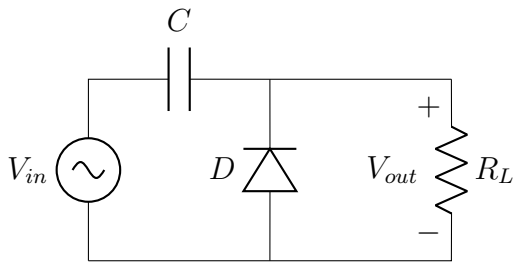


Figure 7.4: Positive Clamper Circuit (Diode points UP).

- Polarity: Left plate (-), Right plate (+).
- $V_{out} \approx 0$ V (Short circuit via diode).

2. **Positive Half Cycle:** The input goes positive.
   - The diode turns OFF (Reverse Bias).
   - The capacitor acts like a battery in series with the source.
   - Kirchhoff's Voltage Law: $V_{out} = V_{in} + V_{capacitor}$.
   - Since $V_C = V_m$, then $V_{out} = V_{in} + V_m$.
3. **Result:** The output swings from 0 to $2V_m$. The DC level is shifted by $+V_m$.

7.3.2 Negative Clamper

This circuit shifts the waveform **DOWNWARDS**. The diode points **DOWN**.

- During the positive half cycle, the capacitor charges to V_m.
- During the negative half cycle, the input and capacitor voltages add up negatively.
- $V_{out} = V_{in} - V_m$.
- The output swings from 0 to $-2V_m$.

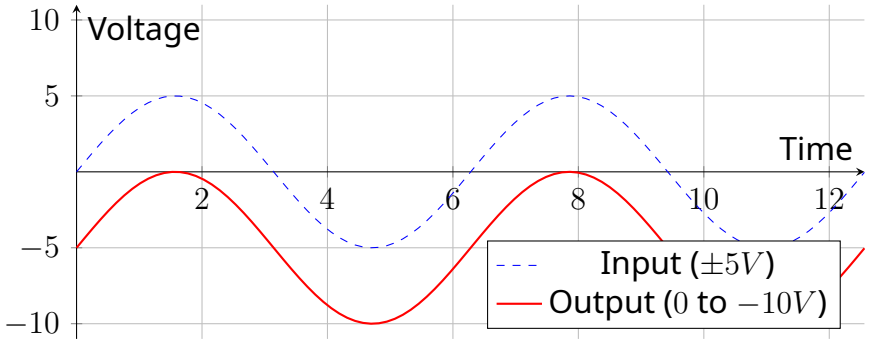


Figure 7.5: Input vs. Output waveforms for a Negative Clamper.

Time Constant Requirement

For a clamper to work effectively, the capacitor must hold its charge during the non-conducting interval. Therefore, the discharge time constant ($\tau = R_L C$) must be large compared to the period of the input signal (T).

$$\tau = R_L C \gg \frac{T}{2}$$

Usually, we design for $\tau \geq 10T$.

7.4 Voltage Multipliers

Voltage multipliers allow us to generate very high DC voltages (e.g., hundreds or thousands of volts) from a lower AC voltage source, without using a heavy step-up transformer. They are commonly used in CRT displays, X-ray tubes, and bug zappers.

7.4.1 Half-Wave Voltage Doubler

This circuit produces a DC output voltage that is **twice** the peak input AC voltage ($2V_m$).

Operation:

1. **Negative Half Cycle:**
   - D_1 is Forward Biased. D_2 is OFF.
   - Capacitor C_1 charges up to V_m.

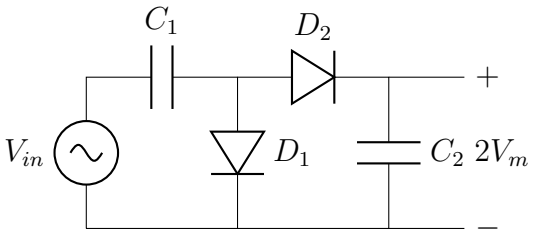


Figure 7.6: Half-Wave Voltage Doubler Circuit.

2. **Positive Half Cycle:**
   - D_1 is OFF. D_2 is Forward Biased.
   - C_1 is already charged to V_m. The source is also at V_m.
   - These two voltages are in series.
   - Capacitor C_2 charges to $V_{in} + V_{C1} = V_m + V_m = 2V_m$.
3. The output across C_2 is a smooth DC voltage of approximately $2V_m$.

7.5 Agricultural Application: Protection Circuits

In agricultural machinery, inductive loads like starter motors and solenoids generate massive negative voltage spikes (inductive kickback) when turned off. These spikes can destroy the ECU transistors.

A **Clipping Diode** (often called a Flyback or Freewheeling Diode) is placed in parallel with the inductive load to clip this spike.

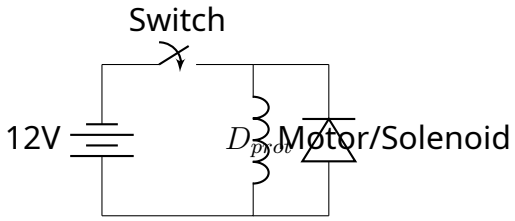


Figure 7.7: Flyback Diode protecting a switch from inductive spikes.

7.6 Review Questions

Part A: Conceptual Questions

1. Define ”Clipping” and ”Clamping”. What is the fundamental difference between their effects on a waveform?
2. Draw the circuit for a Negative Shunt Clipper. Explain its operation during the positive and negative half-cycles.
3. Why is a series resistor required in a Shunt Clipper circuit but not strictly necessary in a Series Clipper?
4. What is the purpose of the battery (V_{bias}) in a Biased Clipper circuit?
5. Explain why a Clamper circuit is sometimes called a ”DC Restorer”.
6. In a positive clamper circuit, which way does the diode point (up or down)?
7. What is the role of the capacitor in a Voltage Doubler circuit?
8. Why is a ”Flyback Diode” used across an inductive load like a motor?
9. How does the ”Barrier Potential” of a silicon diode (0.7V) affect the output of a standard clipper circuit?
10. Explain why the time constant $\tau = RC$ must be large in a clamper circuit.

Part B: Advanced Calculations & Analysis

11. **Biased Clipper Design:** You need to design a circuit to protect a microprocessor input that can tolerate a maximum of 3.3V. The input signal can spike up to 10V. Design a biased shunt clipper using a Silicon diode to limit the voltage to approximately 3.3V. Draw the circuit and calculate the required bias battery voltage.
12. **Clamper Analysis:** A positive clamper has an input square wave of ± 10V (20V peak-to-peak). The capacitor charges to the peak voltage.

(a) What is the maximum positive voltage of the output?

(b) What is the minimum voltage of the output?

(c) What is the average (DC) value of the output waveform?

13. **Voltage Doubler:** A Half-Wave Voltage Doubler is connected to a 120V RMS AC supply.

    (a) Calculate the peak input voltage V_m.

    (b) Calculate the final DC output voltage across the second capacitor.

    (c) What PIV rating must the diodes have?

14. **Waveform Sketching:** An input signal $V_{in} = 10 \sin(\omega t)$ is applied to a biased series clipper. The diode is ideal. The cathode is connected to the source, and the anode is connected to a +4V battery. Sketch the output waveform V_{out} relative to the input.

15. **Time Constant Check:** A clamper circuit operates at a frequency of 1 kHz. The load resistance is $10k\Omega$. Calculate the minimum capacitor value required to ensure the discharge time constant is at least 10 times the signal period.